

Homework 3.

Problem 1:

a) Global HE:

Pixel value	# pixels	r_k	CDF (S_k)	$S_k * 15$	Equalized level
0	0	0	0	0	0
1	0	0	0	0	0
2	6	$1/6$	$1/6$	2.5	3
3	0	0	$1/6$	2.5	3
4	0	0	$1/6$	2.5	3
5	6	$1/6$	$1/3$	5	5
6	0	0	$1/3$	5	5
7	10	$5/18$	$11/18$	9.166	9
8	6	$1/6$	$7/9$	11.66	12
9	8	$2/9$	1	15	15
10	0	0	1	15	15
11	0	0	1	15	15
12	0	0	1	15	15
13	0	0	1	15	15
14	0	0	1	15	15
15	0	0	1	15	15

Equalized Image:

3	3	3	5	5	15
3	3	3	5	5	15
12	12	12	5	5	15
9	9	9	9	9	15
9	9	9	9	9	15
12	12	12	15	15	15

Histogram of the equalized image:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	6	0	6	0	0	0	10	0	0	6	0	0	8

b, Local HE:

①	②
③	④

⊗ Window ①:

2	2	2
2	2	2
8	8	8

pixel value	#pixels	r_k	CDF	Equalized value
2	6	$2/3$	$2/3$	10
8	3	$1/3$	1	15

⊗ Window ② :

$$\begin{bmatrix} 5 & 5 & 9 \\ 5 & 5 & 9 \\ 5 & 5 & 9 \end{bmatrix}$$

pixel value	#pixels	r_k	CDF	Equalized value
5	6	$2/3$	$2/3$	10
9	3	$1/3$	1	15

⊗ Window ③ :

$$\begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \\ 8 & 8 & 8 \end{bmatrix}$$

pixel value	#pixels	r_k	CDF	Equalized value
7	6	$2/3$	$2/3$	10
8	3	$1/3$	1	15

⊗ Window 4:

$$\begin{bmatrix} 7 & 7 & 9 \\ 7 & 7 & 9 \\ 9 & 9 & 9 \end{bmatrix}$$

pixel value	#pixels	r_k	CDF	Equalized value
7	4	$4/9$	$4/9$	7
9	5	$5/9$	1	15

Equalized image:

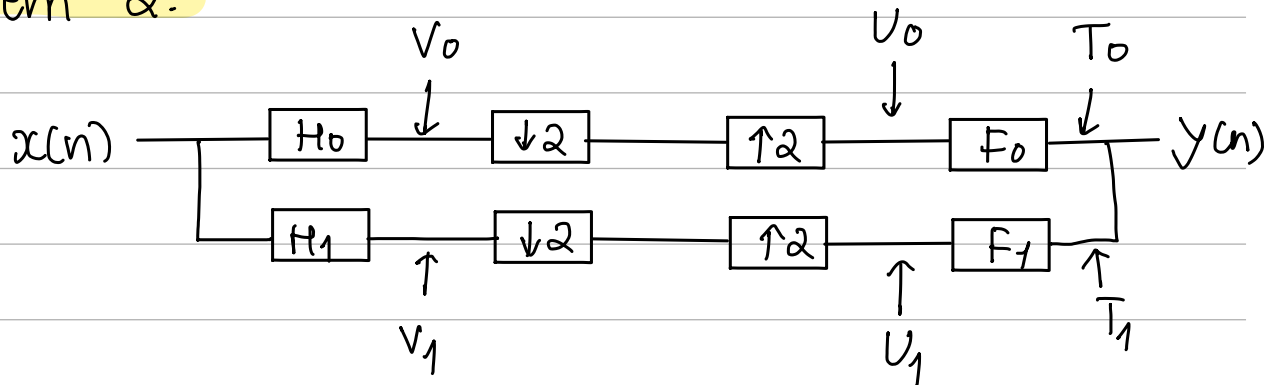
$$\begin{bmatrix} 10 & 10 & 10 & 10 & 10 & 15 \\ 10 & 10 & 10 & 10 & 10 & 15 \\ 15 & 15 & 15 & 10 & 10 & 15 \\ 10 & 10 & 10 & 7 & 7 & 15 \\ 10 & 10 & 10 & 7 & 7 & 15 \\ 15 & 15 & 15 & 15 & 15 & 15 \end{bmatrix}$$

c) Comparison:

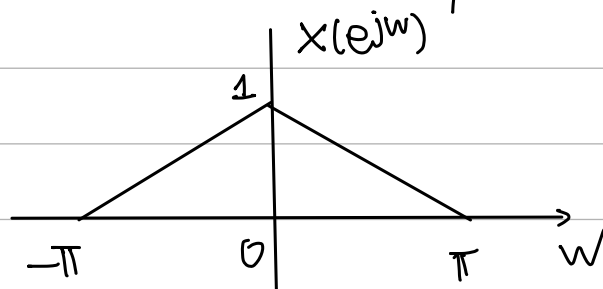
Local HE enhances contrast in local regions, amplifying the edges.

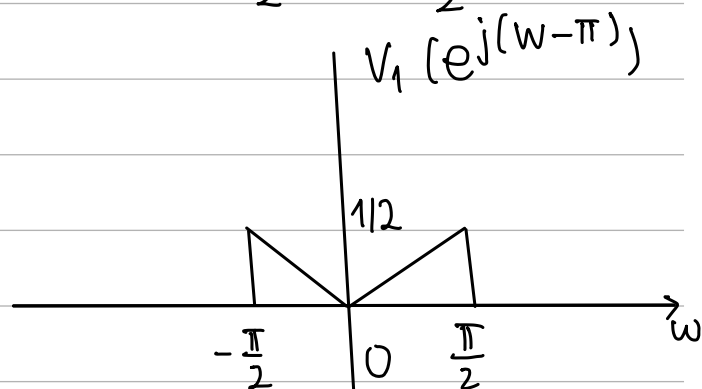
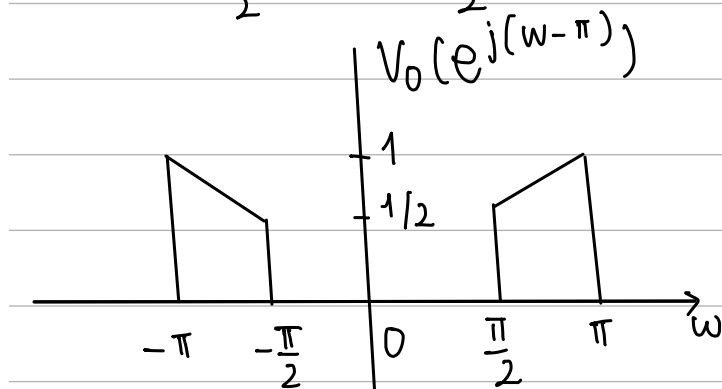
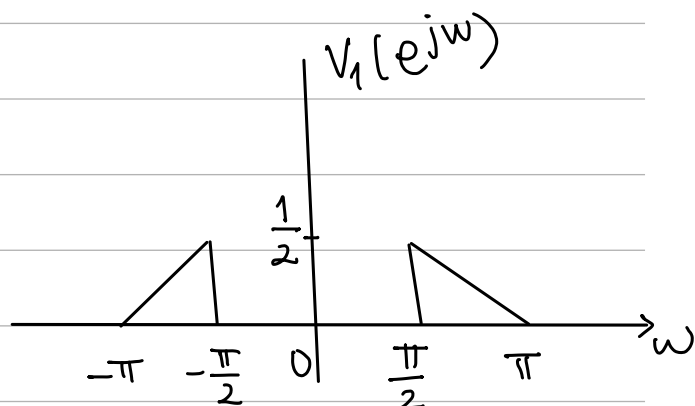
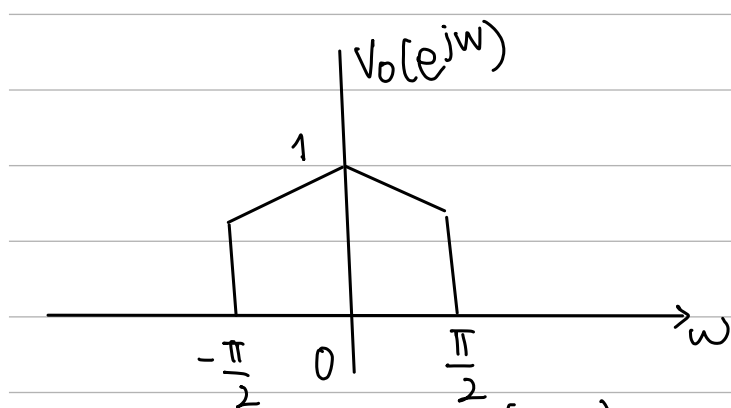
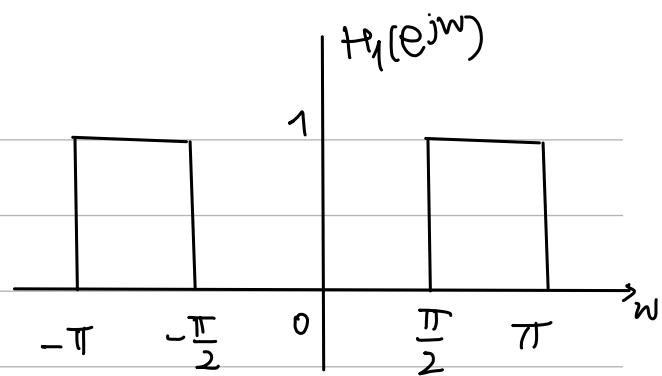
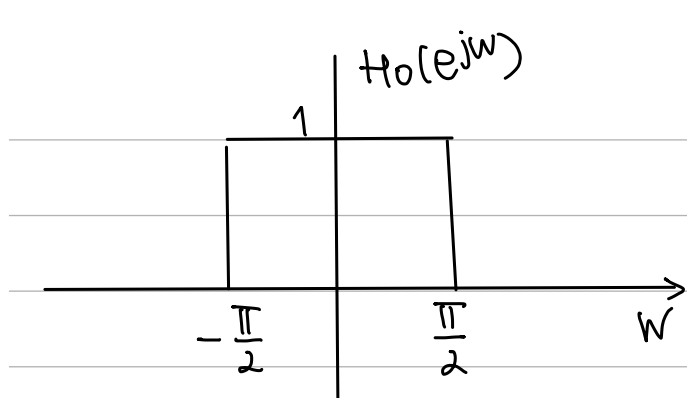
Problem 2:

a)

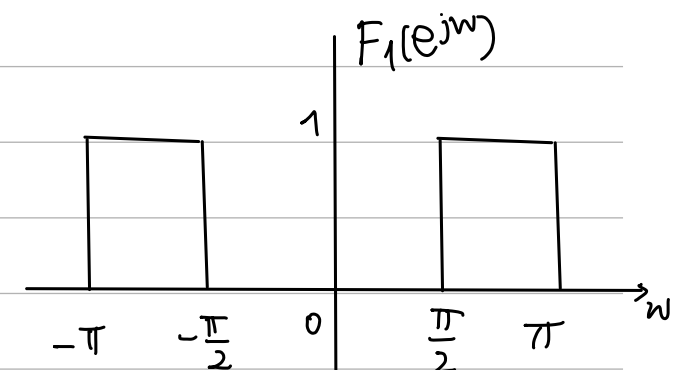
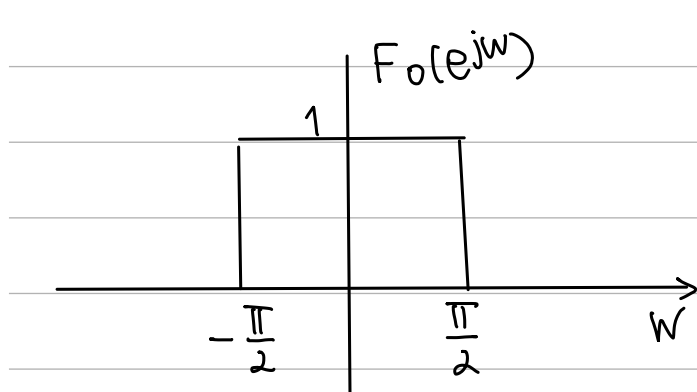
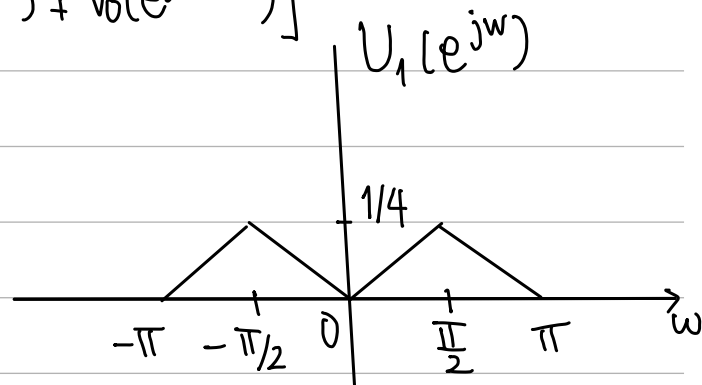
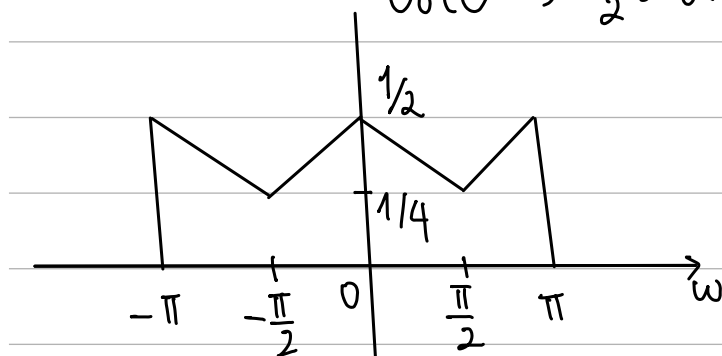


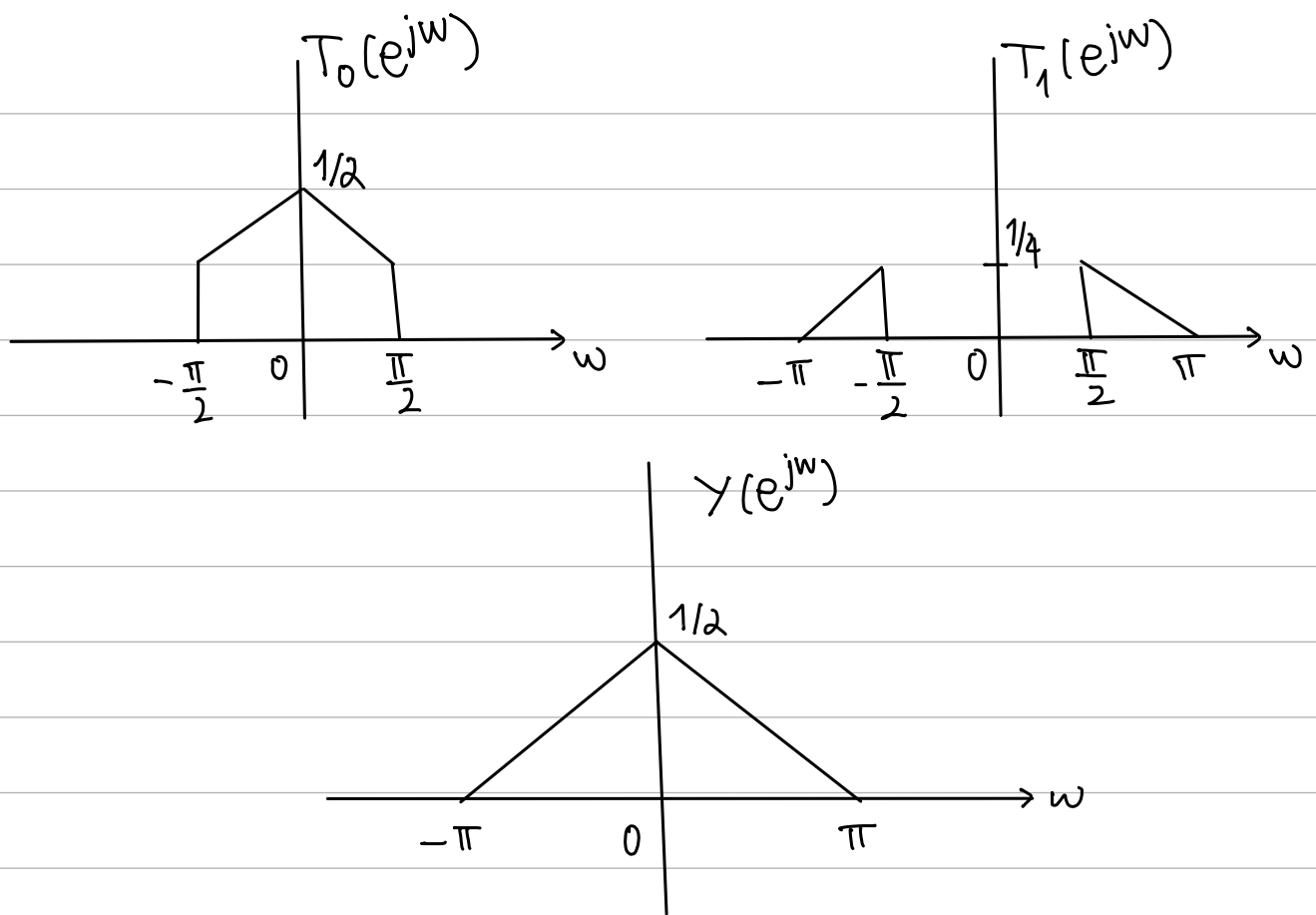
Let $X(e^{j\omega})$ has a spectrum:





$$U_0(e^{j\omega}) = \frac{1}{2} [V_0(e^{j\omega}) + V_0(e^{j(\omega-\pi)})]$$





Output has the spectrum with the input, and there is no aliasing.

$$b) \quad H_0(z) = \frac{1}{\sqrt{2}} (1 + \bar{z}^1) \quad , \quad H_1(z) = \frac{1}{\sqrt{2}} (1 - \bar{z}^1)$$

$$F_0(z) = H_1(-z) = \frac{1}{\sqrt{2}} (1 + \bar{z}^1)$$

$$F_1(z) = -H_0(-z) = -\frac{1}{\sqrt{2}} (1 - \bar{z}^1)$$

⊗ Verify PR:

$$\begin{aligned} & F_0(z)H_0(z) + F_1(z)H_1(z) \\ &= \frac{1}{\sqrt{2}} (1 + \bar{z}^1) \frac{1}{\sqrt{2}} (1 + \bar{z}^1) - \frac{1}{\sqrt{2}} (1 - \bar{z}^1) \frac{1}{\sqrt{2}} (1 - \bar{z}^1) \\ &= \frac{1}{2} (1 + 2\bar{z}^1 + \bar{z}^2) - \frac{1}{2} (1 - 2\bar{z}^1 + \bar{z}^2) \end{aligned}$$

$$= 2\bar{z}^1$$

$$\begin{aligned}
 & F_0(z)H_0(-z) + F_1(z)H_1(-z) \\
 = & \frac{1}{\sqrt{2}}(1+\bar{z}^1) \frac{1}{\sqrt{2}}(1-\bar{z}^1) - \frac{1}{\sqrt{2}}(1-\bar{z}^1) \frac{1}{\sqrt{2}}(1+\bar{z}^1) \\
 = & \frac{1}{2}(1-\bar{z}^2) - \frac{1}{2}(1-\bar{z}^2) \\
 = & 0
 \end{aligned}$$

Therefore, overall system is PR.

Problem 3:

a) $F_0(z) = H_0(z)$

Addition, $F_1(z) = -H_0(-z)$

$$H_1(z) = F_0(-z) = H_0(-z)$$

$\Rightarrow$ PR condition in terms of $H_0(z)$:

$$\begin{cases}
 H_0(z)H_0(z) - H_0(-z)H_0(-z) = 2\bar{z}^L \\
 H_0(z)H_0(-z) - H_0(-z)H_0(z) = 0
 \end{cases}$$

$$\Rightarrow H_0^2(z) - H_0^2(-z) = 2\bar{z}^L$$

$$b, \quad H_0^2(z) - H_0^2(-z) = 2\bar{z}^L = 2\bar{z}^{(2k+1)}$$

$$(H_0(z) + H_0(-z))(H_0(z) - H_0(-z)) = 2\bar{z}^{(2k+1)}$$

$$\Rightarrow \begin{cases} H_0(z) + H_0(-z) = \alpha_0 \bar{z}^{\beta_0} \\ H_0(z) - H_0(-z) = \alpha_1 \bar{z}^{\beta_1} \end{cases}$$

$$\text{where } \alpha_0 \alpha_1 = 2 \quad \text{and} \quad \beta_0 + \beta_1 = 2k+1$$

$$\Rightarrow H_0(z) = \frac{1}{2} (\alpha_0 \bar{z}^{\beta_0} + \alpha_1 \bar{z}^{\beta_1})$$

(weighted sum of 2 delays)

$$c, \quad H_1(z) = H_0(-z) = \frac{1}{2} [\alpha_0 (-z)^{-\beta_0} + \alpha_1 (-z)^{\beta_1}]$$

$$F_0(z) = H_0(z) = \frac{1}{2} (\alpha_0 \bar{z}^{\beta_0} + \alpha_1 \bar{z}^{\beta_1})$$

$$F_1(z) = -H_0(-z) = -\frac{1}{2} [\alpha_0 (-z)^{-\beta_0} + \alpha_1 (-z)^{\beta_1}]$$